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M.Sc. in Physics and Mathematics (Fisymat)

Numerical Analysis of PDEs and Approximation

Chebyshev Spectral–Galerkin Method for the Helmholtz Equation

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Abstract

We derive and implement a Chebyshev spectral–Galerkin method for the two-dimensional Helmholtz equation with homogeneous Dirichlet boundary conditions on the square $\Omega = (-1,1)^2$. A boundary-adapted basis $\phi_n(x) = T_{n+2}(x) - T_n(x)$ is combined with a tensor-product construction on Ω, and the Galerkin system is assembled through Kronecker products of 1D mass and stiffness matrices. Against a smooth manufactured solution we observe spectral (essentially exponential) convergence down to machine precision, a polynomial growth of the condition number $\kappa(A) \sim N^4$ for the 2D Laplacian, and a systematic improvement of both accuracy and conditioning as the reaction coefficient k increases.

1 Problem statement

We consider the Helmholtz equation on the square $\Omega = (-1,1) \times (-1,1)$,

$$-\Delta u(x,y) + k^2 u(x,y) = f(x,y), \qquad (x,y) \in \Omega, \tag{1.1}$$

subject to homogeneous Dirichlet boundary conditions

$$u(x,y) = 0, \qquad (x,y) \in \partial\Omega, \tag{1.2}$$

with a given wavenumber $k > 0$.

In order to validate the numerical method we use the method of manufactured solutions. We prescribe

$$u_{\text{exact}}(x,y) = (1-x^2)(1-y^2)\, e^{-x-y}, \tag{1.3}$$

which vanishes on $\partial\Omega$ by construction, and then compute the source term f analytically so that (1.3) is the exact solution of (1.1).

1.1 Source term for the manufactured solution

Let $h(x) = (1-x^2)e^{-x}$ and note that $u = h(x)\, h_y(y)$ with $h_y(y) = (1-y^2)e^{-y}$. A direct computation gives

$$\begin{aligned} h'(x) &= e^{-x}\big(x^2 - 2x - 1\big), \\ h''(x) &= e^{-x}\big(-x^2 + 4x - 1\big), \end{aligned}$$

so that $u_{xx} = h''(x)(1-y^2)e^{-y}$ and likewise for u_{yy}. Collecting terms,

$$\boxed{f(x,y) = e^{-x-y}\Big[(1-y^2)(x^2-4x+1) + (1-x^2)(y^2-4y+1) + k^2(1-x^2)(1-y^2)\Big].} \tag{1.4}$$

2 Weak formulation

Let $V = H_0^1(\Omega)$. Multiplying (1.1) by a test function $v \in V$, integrating on Ω and applying Green's identity with $v|_{\partial\Omega} = 0$ yields the variational problem: find $u \in V$ such that

$$a(u,v) = L(v) \qquad \forall\, v \in V, \tag{2.1}$$

where

$$a(u,v) = \int_\Omega \nabla u \cdot \nabla v \, dxdy + k^2 \int_\Omega u\, v \, dxdy, \tag{2.2}$$

$$L(v) = \int_\Omega f\, v \, dxdy. \tag{2.3}$$

The bilinear form $a(\cdot,\cdot)$ is continuous and coercive on V: for all $u \in V$,

$$a(u,u) \;\geq\; \min(1,k^2)\,\|u\|^2_{H^1_0(\Omega)}.$$

Hence the Lax–Milgram theorem [3] yields a unique weak solution.

3 Spectral-Galerkin discretisation

3.1 Boundary-adapted basis

We introduce the Chebyshev-based basis

$$\phi_n(x) = T_{n+2}(x) - T_n(x), \qquad n = 0,1,\ldots,N-1, \tag{3.1}$$

where T_n is the Chebyshev polynomial of the first kind of degree n. Since $T_m(\pm 1) = (\pm 1)^m$, every ϕ_n satisfies

$$\phi_n(\pm 1) = 0,$$

so the Dirichlet condition is built into the basis. We then take the finite-dimensional subspace

$$V_N = \operatorname{span}\{\phi_i(x)\,\phi_j(y) : 0 \leq i,j \leq N-1\} \subset V.$$

3.2 Galerkin approximation

The discrete solution is sought as

$$u_N(x,y) = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1} u_{ij}\,\phi_i(x)\,\phi_j(y), \tag{3.2}$$

and imposing (2.1) for every test function $\phi_m(x)\phi_n(y)$ gives

$$\sum_{i,j} u_{ij}\,a\big(\phi_i\phi_j, \phi_m\phi_n\big) = L\big(\phi_m\phi_n\big). \tag{3.3}$$

3.3 Kronecker-product assembly

Because the basis is a tensor product, the 2D integrals factorise into products of 1D integrals:

$$\int_\Omega \nabla(\phi_i\phi_j)\cdot\nabla(\phi_m\phi_n)\,dxdy = K_{im}\,M_{jn} + M_{im}\,K_{jn},$$
$$\int_\Omega \phi_i\phi_j\phi_m\phi_n\,dxdy = M_{im}\,M_{jn},$$

where

$$M_{ij} = \int_{-1}^{1} \phi_i(x)\phi_j(x)\,dx, \qquad K_{ij} = \int_{-1}^{1} \phi_i'(x)\phi_j'(x)\,dx. \tag{3.4}$$

Flattening the coefficient matrix u_{ij} row-major into a vector $\mathbf{u} \in \mathbb{R}^{N^2}$, the Galerkin system takes the compact Kronecker form

$$\boxed{A\,\mathbf{u} = \mathbf{b}, \qquad A = K \otimes M + M \otimes K + k^2\,(M \otimes M), \qquad b_{mn} = \int_\Omega f\,\phi_m\phi_n\,dxdy.} \tag{3.5}$$

For a given N, assembling A and $\mathbf{b}$ costs $\mathcal{O}(N^3)$, and solving (3.5) costs $\mathcal{O}(N^6)$ with a direct dense solver, or less if the Kronecker structure is exploited.

3.4 Quadrature

The 1D integrals defining M and K involve polynomials of degree up to $2(N+1)$; they are computed *exactly* (up to floating point) with a Gauss–Legendre rule of $N_q \geq N+2$ nodes [2]. The right-hand side $\mathbf{b}$ uses the same Gauss–Legendre nodes tensorised on Ω, yielding the compact formula

$$b_{mn} = \sum_{p=1}^{N_q}\sum_{q=1}^{N_q} w_p\,w_q\,f(x_p,y_q)\,\phi_m(x_p)\,\phi_n(y_q).$$

4 Python implementation

The core of the method fits in a single module `chebyshev_core.py`. Once the basis tables on the quadrature nodes are assembled, M and K are obtained as two BLAS-3 multiplications and the right-hand side as a single $\Phi\,F\,\Phi^T$ contraction.

```
import numpy as np

def chebyshev_T(n, x):
    return np.cos(n * np.arccos(np.clip(x, -1.0, 1.0)))

def phi(n, x):
    return chebyshev_T(n+2, x) - chebyshev_T(n, x)

def gauss_legendre_quadrature(Nq):
    x, w = np.polynomial.legendre.leggauss(Nq)
    return x, w

def build_1D_matrices(N, Nq):
    x, w = gauss_legendre_quadrature(Nq)
    Phi  = np.vstack([phi(n, x)              for n in range(N)])
    dPhi = np.vstack([phi_derivative(n, x)   for n in range(N)])
    M = (Phi  * w) @ Phi.T
    K = (dPhi * w) @ dPhi.T
    return M, K

def build_2D_matrix(M, K, k):
    return np.kron(K, M) + np.kron(M, K) + k**2 * np.kron(M, M)
```

```
def build_rhs(N, Nq, k):
    x, w = gauss_legendre_quadrature(Nq)
    Phi  = np.vstack([phi(n, x) for n in range(N)])
    X, Y = np.meshgrid(x, x, indexing="ij")
    F    = f_source(X, Y, k)
    return ((Phi * w) @ F @ (Phi * w).T).flatten()
```

Listing 1: Core assembly routines.

5 Numerical results

5.1 Solution for $N = 12$, $k = 5$

For the manufactured problem, already $N = 12$ basis functions per direction recover the solution to $\sim 10^{-13}$ in the maximum norm.

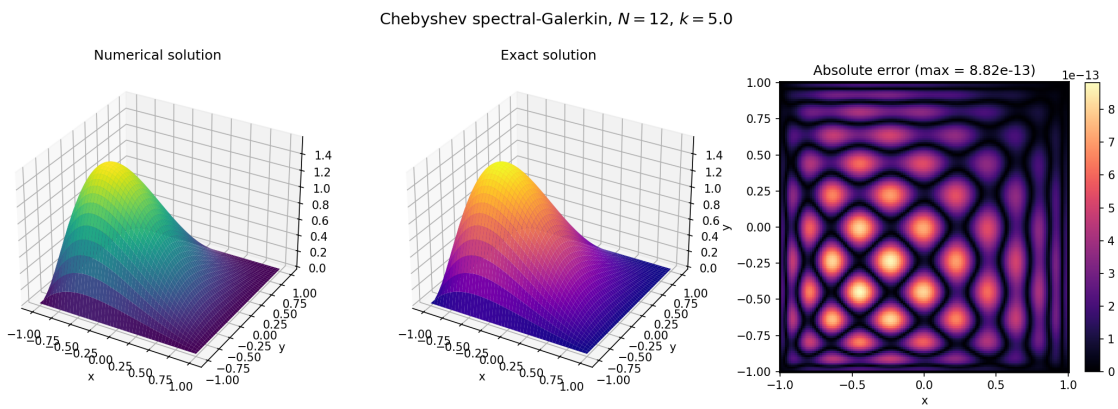

Figure 1: Numerical solution u_N (left), exact solution u_{exact} (middle) and pointwise absolute error (right) for $N = 12$, $N_q = 40$, $k = 5$.

5.2 Spectral convergence

For the smooth C^∞ manufactured solution the Chebyshev spectral-Galerkin error decays faster than any polynomial rate, saturating at floating-point precision.

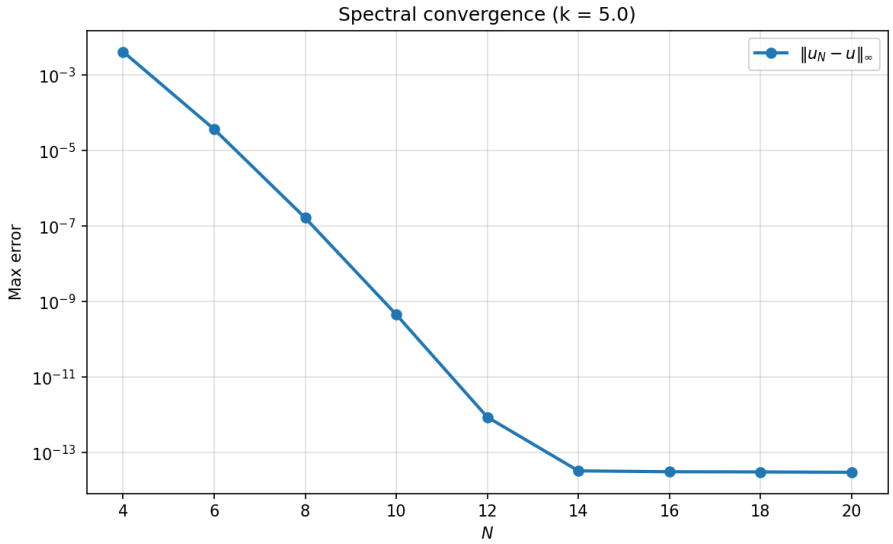

Figure 2: Maximum error as a function of the basis size N, in log scale. The error drops from $\sim 10^{-3}$ at $N = 4$ to the $\mathcal{O}(10^{-14})$ floor at $N \geq 14$, reflecting spectral (essentially exponential) convergence.

5.3 Condition number

The 2D stiffness matrix A is dense and becomes increasingly ill-conditioned as N grows. For the Chebyshev boundary-adapted basis $\phi_n = T_{n+2} - T_n$ the 1D stiffness eigenvalues grow like $\mathcal{O}(n^4)$ [4], and the tensor structure of (3.5) propagates the same $\mathcal{O}(N^4)$ rate to the 2D operator.

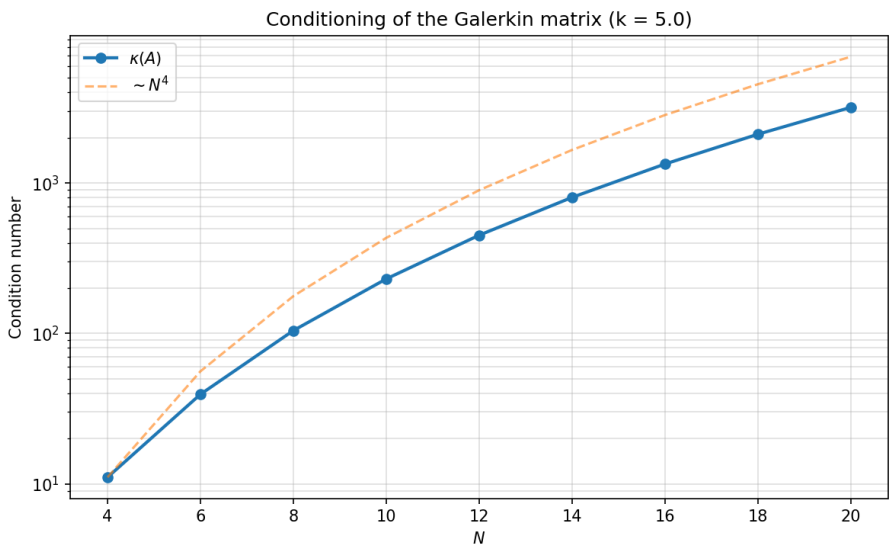

Figure 3: Condition number $\kappa(A)$ as a function of N for $k = 5$. The dashed line shows the reference slope $\kappa \sim N^4$, which matches the measured growth closely.

5.4 Influence of the wavenumber k

For this particular smooth non-oscillatory manufactured solution, increasing the reaction coefficient k reinforces the coercivity of $a(\cdot,\cdot)$ and pushes the mass contribution of (3.5) towards dominance. Both the error *and* the condition number therefore *decrease* with k.

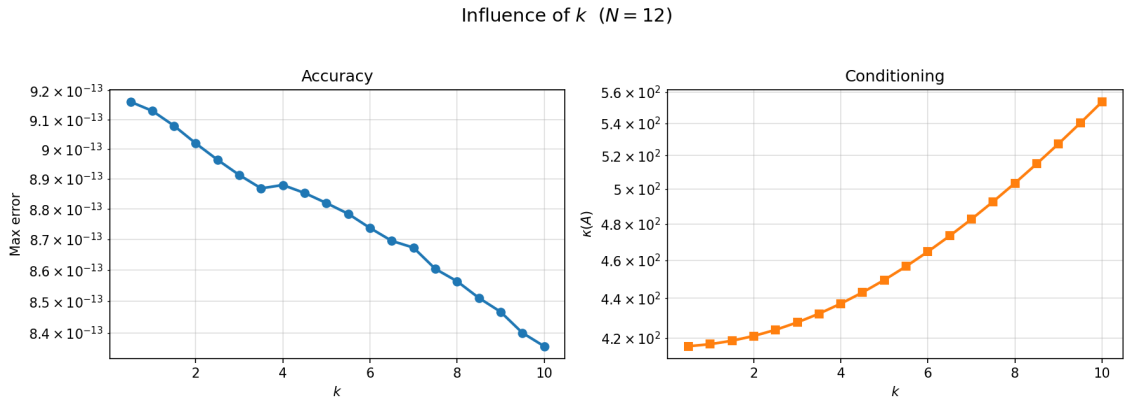

Figure 4: Accuracy (left) and conditioning (right) as a function of k for $N = 12$. The error drops by about one order of magnitude as k goes from 0.5 to 10; the condition number decreases by roughly the same factor.

6 Conclusions

A Chebyshev spectral–Galerkin method has been derived and implemented for the 2D Helmholtz equation with Dirichlet boundary conditions. The boundary-adapted basis $\phi_n = T_{n+2} - T_n$ enforces the boundary conditions exactly, and the tensor-product construction reduces the 2D assembly to three Kronecker products of 1D matrices, each computed exactly by Gauss–Legendre quadrature. On a smooth manufactured solution we observed

- **spectral convergence** of the maximum error down to machine precision for modest sizes $N \geq 14$;
- an $\mathcal{O}(N^4)$ growth of the condition number of the 2D Galerkin matrix, consistent with classical bounds for Chebyshev methods [4];
- a mild *improvement* of both accuracy and conditioning with the reaction coefficient k, caused by the increased coercivity of the bilinear form.

These results are qualitatively in line with the literature on spectral methods for elliptic PDEs [1, 5, 2]. The inherent ill-conditioning is the main practical limitation of the approach in double precision, although it is acceptable in this project because the error saturates at 10^{-13} well before $\kappa(A)$ becomes problematic.

References

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